

Interactive Data Visualization Using Plotly



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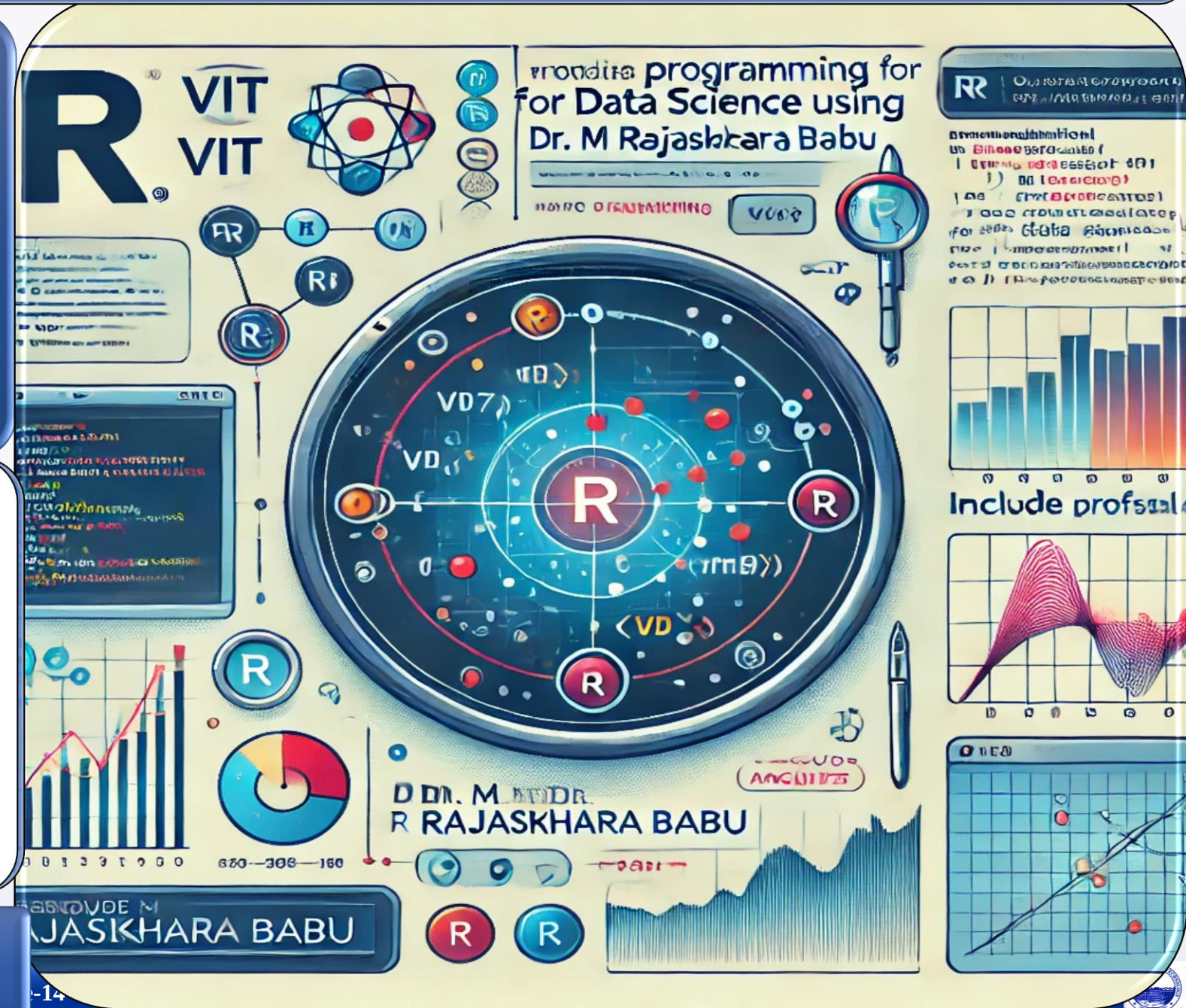


VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Vellore-632014, Tamil Nadu, India



What is Data Visualization?

- Data Visualization is the graphical representation of data to make information easier to understand and analyze.
- It transforms raw numbers into visual stories that reveal patterns, trends, and insights at a glance.



Easy Interpretation
Complex data made simple



Trend Identification
Spot patterns instantly



Better Decisions
Data-driven choices



Interactive Exploration
Engage with your data

What is Plotly?

Plotly is a powerful Python visualization library used to create interactive charts, dashboards, scientific visualizations, and business reports — all rendered in the browser.

Interactive Graphs

Click, hover, and explore data dynamically

Zoom & Pan

Navigate large datasets with ease

Hover Information

Reveal details on mouse-over

Export Charts

Save as image or embed in web apps



Install Plotly via pip, then import the required libraries to begin building interactive visualizations.

Install Plotly

```
pip install plotly
```

Import Libraries

```
import plotly.express as px  
import pandas as pd
```

 Both `plotly` and `pandas` are required for this project. Make sure both are installed before proceeding.

Create an interactive bar chart showing the population of different cities.

Sample Dataset

City	Population (Lakhs)
Mumbai	204
Delhi	167
Bengaluru	136
Hyderabad	108
Chennai	115
Kolkata	149

Creating a DataFrame

We use `pandas` to structure our city population data into a `DataFrame` — the standard tabular format used in Python data science workflows.

```
import pandas as pd

data = {
    'City': ['Mumbai', 'Delhi', 'Bengaluru',
            'Hyderabad', 'Chennai', 'Kolkata'],
    'Population': [204, 167, 136, 108, 115, 149]
}

df = pd.DataFrame(data)
print(df)
```

- The `pd.DataFrame()` constructor converts a Python dictionary into a structured table with labeled rows and columns.

What is a Bar Chart?

A bar chart displays categorical data using rectangular bars, where bar height represents the value for each category.

Components

- Categories → Cities (X-axis)
- Values → Population (Y-axis)

Common Applications

- **Sales Analysis**
- **Population Studies**
- **Market Research**
- **Survey Results**

Creating an Interactive Bar Chart

Use `plotly.express.bar()` to generate an interactive chart in just a few lines. The `fig.show()` call renders it in your browser.

```
import plotly.express as px

fig = px.bar(
    df,
    x='City',
    y='Population',
    title='Population of Different Cities'
)

fig.show()
```

- ✓ Running this code opens an interactive chart in your default web browser — no additional configuration needed!

Interactive Features of Plotly



Hover Information

Exact values appear when you hover over any bar



Zoom In / Zoom Out

Focus on specific data ranges with scroll or drag



Pan

Navigate across the chart by clicking and dragging



Save as Image

Export the chart as a PNG with one click

Adding Colors to Bars

Use the `color` parameter to map bar colors to population values. This creates a color gradient that makes it visually easy to compare cities at a glance.

```
fig = px.bar(  
    df,  
    x='City',  
    y='Population',  
    color='Population',  
    title='City Population Analysis'  
)  
  
fig.show()
```

- ❑ Different populations are represented using different colors — higher values appear in a distinct shade, reinforcing the visual hierarchy.

Custom Axis Labels

```
fig = px.bar(  
    df,  
    x='City',  
    y='Population',  
    labels={  
        'Population':  
        'Population (Lakhs)'  
    },  
    title='Population  
Distribution'  
)
```

Display Values on Bars

```
fig = px.bar(  
    df,  
    x='City',  
    y='Population',  
    text='Population'  
)  
fig.show()
```

The `text='Population'` parameter places population values above each bar for immediate readability.

Complete Python Program

The full program combines all customizations: color mapping, value labels, and axis titles into one polished, interactive chart.

```
import pandas as pd
import plotly.express as px

data = {
    'City': ['Mumbai', 'Delhi', 'Bengaluru',
             'Hyderabad', 'Chennai', 'Kolkata'],
    'Population': [204, 167, 136, 108, 115, 149]
}
df = pd.DataFrame(data)

fig = px.bar(df, x='City', y='Population',
             color='Population', text='Population',
             title='Population of Different Cities')

fig.update_layout(xaxis_title='City',
                  yaxis_title='Population (Lakhs)')
fig.show()
```

Sample Output Interpretation

Running the complete program produces a color-coded interactive bar chart. Here are the key observations from the output:

Highest Population

Mumbai leads with 204 Lakhs — the tallest bar in the chart.

Lowest Population

Hyderabad has the lowest at 108 Lakhs among the listed cities.

Mid-Range Cities

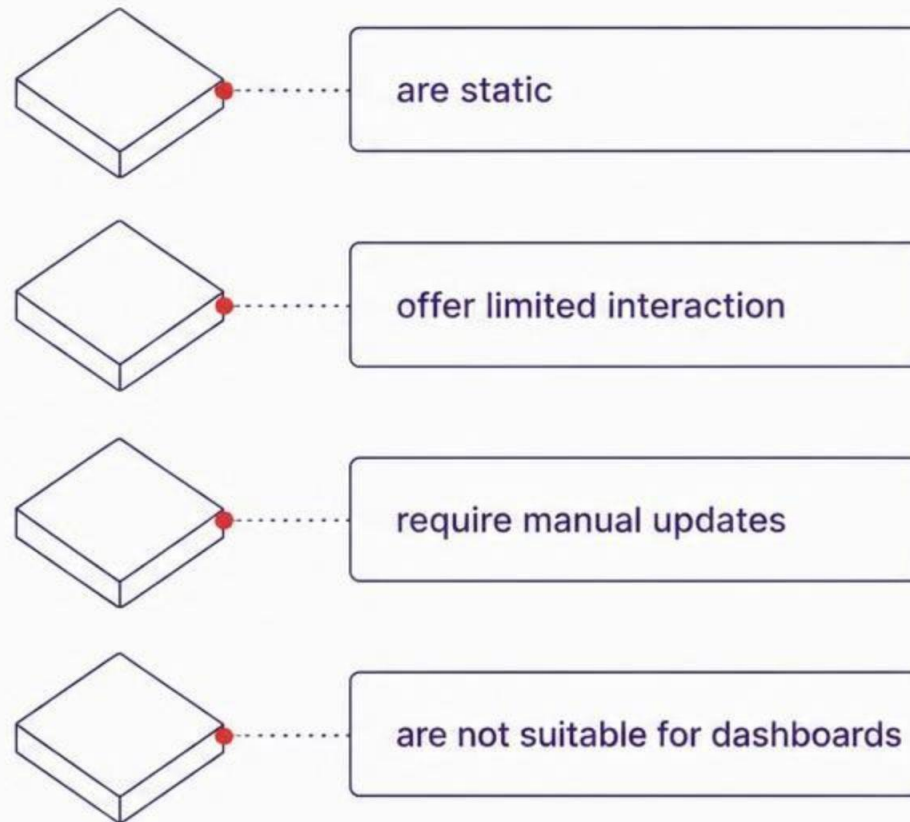
Delhi and Kolkata have relatively high populations at 167 and 149 Lakhs.

Population Ranking

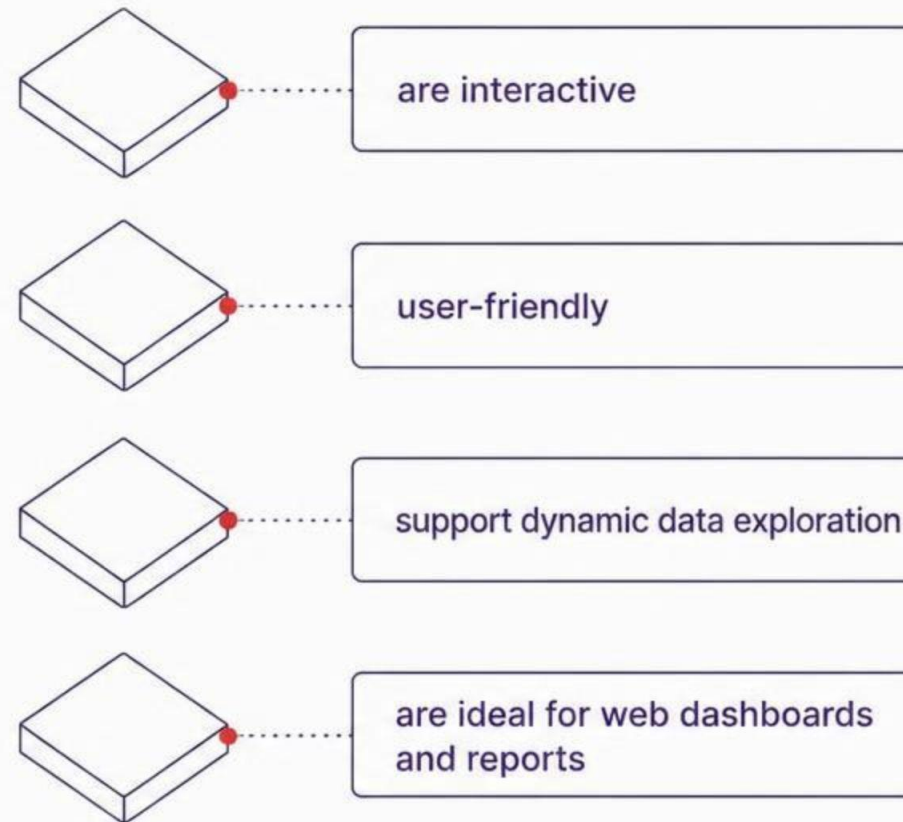
Mumbai > Delhi > Kolkata > Bengaluru > Chennai > Hyderabad

Plotly vs. Traditional Charts

Traditional Charts



Plotly Charts



Plotly's interactivity makes it the preferred choice for modern data science workflows, dashboards, and presentations where user engagement matters.



Government

Population analysis and census reports



Business

Market size analysis and sales tracking



Healthcare

Patient distribution across regions



Smart Cities

Urban planning and resource allocation



Education

Student population analysis by institution

Practice Exercise

Apply what you've learned by creating an interactive bar chart for the following product sales dataset:

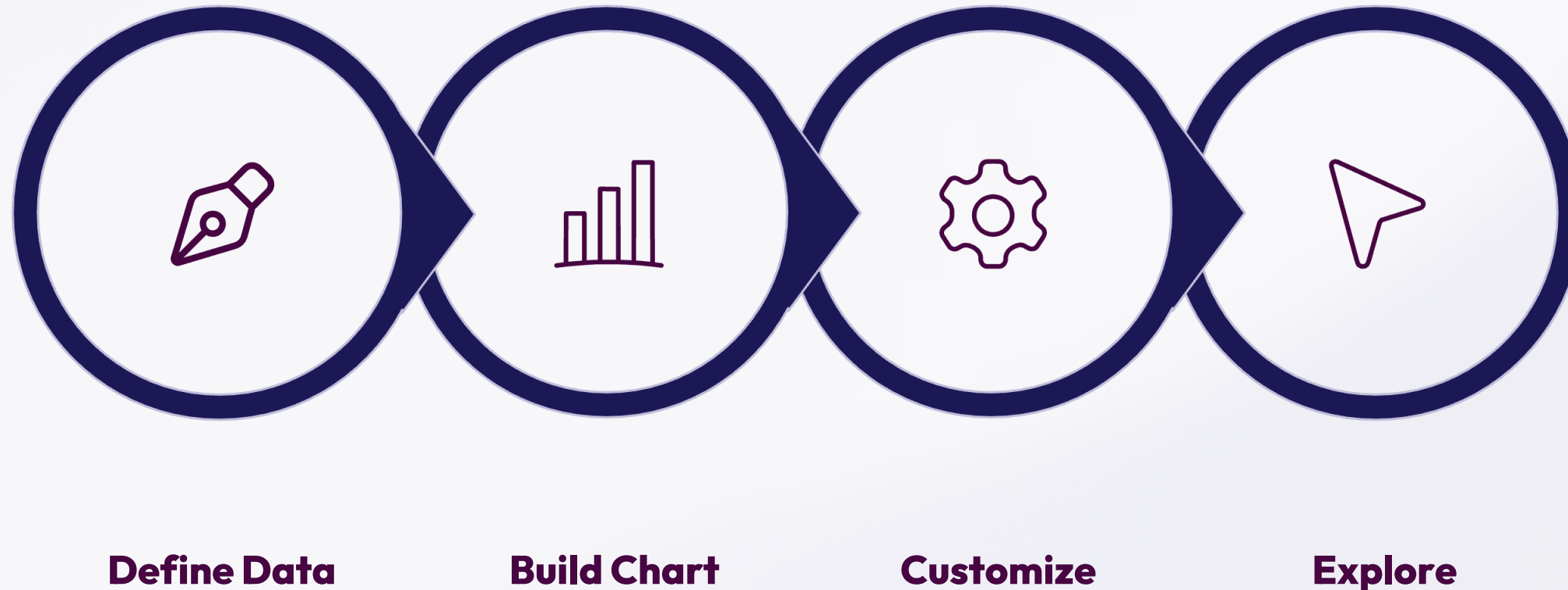
Dataset

Product	Sales
Laptop	450
Mobile	700
Tablet	300
Printer	200

Tasks

1. Create a DataFrame from the dataset
2. Generate a Plotly bar chart
3. Display sales values on bars
4. Add colors and a descriptive title

Key Concepts Recap



This workflow — from raw data to a fully interactive, customized Plotly chart — is the foundation of data visualization in Python data science projects.



Summary: Topics Covered

01

Data Visualization

Graphical representation of data for insight and analysis

02

Plotly Library

Interactive, web-based Python visualization tool

03

DataFrames

Structuring data with pandas for chart input

04

Bar Charts & Customization

Colors, labels, text values, and axis titles

05

Interactive Features

Hover, zoom, pan, and export for dynamic exploration

06

Population Analysis

Real-world application using Indian city data



City Population at a Glance

204

Mumbai

Highest population (Lakhs)

167

Delhi

Second largest city

149

Kolkata

Third in ranking

108

Hyderabad

Lowest in dataset

These figures highlight the uneven population distribution across major Indian cities — a pattern that Plotly's interactive bar chart makes immediately visible and explorable.

Start Visualizing with Plotly

You now have all the tools to build interactive, professional-grade bar charts using Python, pandas, and Plotly. Apply these skills to any dataset — from city populations to product sales — and bring your data to life.

Install

```
pip install plotly
```

Build

Use `px.bar()` with your DataFrame

Explore

Hover, zoom, and export your chart

